

## **Environmental Monitoring and Enhancement of EIA Laboratory at Polar Research Station “Maitri” in Antarctica**

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### **ABSTRACT**

Environmental monitoring was carried out for air and water component during the austral summer of 2003-2004 at the Indian Polar Research Station “Maitri” in compliance with the statutory requirements of the Protocol on Environmental Protection to Antarctica Treaty. The objectives zeroed in on ambient air and water quality assessments, performance evaluation of the Rotating Biological Contactors (RBC) in the wastewater treatment units, enhancement of the Environmental Impact Assessment (EIA) laboratory and its relocation to the old Green House (room) inside the Maitri station. For relocation of the lab basic infrastructure including plumbing for water inlet/outlet, proper electrical outlets, fuming chamber and exhaust arrangement (to vent out gases generated during experiments) was provided in the room. Reagents, glassware, sampling equipments and analytical instruments were arranged in designated areas. The laboratory now has the capacity to carry out a wide range of analysis. For the efficient running of the RBCs in the wastewater units at the designated organic load, it is extremely important and critical to maintain an optimal range of temperature. Good environmental practices being a norm at Maitri station the pollution level is kept (at bay) within permissible level.

### **INTRODUCTION**

Scientific experiment involves human footprints which would lead to contamination, particularly in cold regions and this has only recently become a widely recognized problem. Establishment of infrastructure support to conduct scientific research becomes source of contamination especially in Antarctica. Contaminants typically become mobile in the summer months and are readily dispersed from their immediate sources, where they then cause environmental impacts or pose risks to human health. However, the ways by which contaminants in freezing ground disperse and interact with associated ecosystems is a new and challenging field of applied research.

Such research needs to underpin the design and optimization of techniques for site management and assessment, particularly since remediation in cold regions is inherently more difficult and expensive, than elsewhere.

The Protocol on Environmental Protection of the Antarctic Treaty (Madrid, 1991), was ratified in 1998, does commit Treaty Parties to the comprehensive protection of the Antarctic environment and dependent associated ecosystems, and thus designates Antarctica as a natural reserve devoted to peace and science. Annex III to the protocol (Waste Disposal and Management) established that past and present work sites shall be cleaned up unless they are designated as a historic site or monument, or removal by any practical option would result in greater adverse environmental impact than leaving in its existing location or condition. Some relevant articles of Madrid Protocol are cited below;

***Article 4 (Other Waste disposal on Land)***

1. Wastes not removed or disposed of in accordance with Articles 2 and 3 shall not be disposed of onto ice-free areas or into fresh water systems.
2. Sewage, domestic liquid wastes and other liquid wastes not removed from the Antarctic Treaty Area in accordance with Article 2, shall, to the maximum extent practicable, not be disposed of onto sea ice, ice shelves or the grounded ice-sheet, provided that such wastes which are generated by stations located inland on ice shelves or on the grounded ice-sheet may be disposed of in deep ice pits where such disposal is the only practicable option. Such pits shall not be located on known ice-flow lines which terminate at ice-free areas or in areas of high ablation.
3. Wastes generated at field camps shall, to the maximum extent practicable, be removed by the generator of such wastes to supporting stations or ships for disposal in accordance with this Annex.

***Article 5 (Disposal of Waste in the Sea)***

1. Sewage and domestic liquid wastes may be discharged directly into the sea, taking into account the assimilative capacity of the receiving marine environment and provided that:
  - (a) such discharge is located, wherever practicable, where conditions exist for initial dilution and rapid dispersal; and

- (b) large quantities of such wastes (generated in a station where the average weekly occupancy over the austral summer is approximately 30 individuals or more) shall be treated at least by maceration.
- 2. The by-product of sewage treatment by the Rotary Biological Contactor process or similar processes may be disposed of into the sea provided that such disposal does not adversely affect the local environment, and provided also that any such disposal at sea shall be in accordance with Annex IV to the Protocol.

Annex IV (article 6) to the Protocol (Prevention of Marine Pollution) states that -

- (a) each Party shall eliminate all discharge into the sea of untreated sewage ("sewage" being defined in Annex IV of MARPOL 73/78) within 12 nautical miles of land or ice shelves;
- (b) beyond such distance, sewage stored in a holding tank shall not be discharged instantaneously but at a moderate rate and, where practicable, while the ship is en route at a speed of no less than 4 knots.

## **ENHANCEMENT OF ENVIRONMENTAL LABORATORY AT MAITRI**

Some basic physico-chemical parameters are necessary to be measured at the time of collection of water samples. It was felt necessary to enhance an environmental impact assessment laboratory, which was established during 19<sup>th</sup> ISEA by National Environmental Engineering Research Institute, Nagpur, in a module located in front of Maitri station. It is also decided to relocate the laboratory inside the Maitri station with the capacity to analyze physico-chemical parameters of water and wastewater samples. To set-up an environmental laboratory a suitable place was needed to carry out continuous environmental monitoring and analysis so as to take effective decision and for waste management of Maitri station. For this purpose the old Green House room was selected in Maitri station (Fig. 1). For relocation of the lab basic infrastructure including plumbing for water inlet/outlet, proper electrical outlets, fuming chamber and exhaust arrangement (to vent out gases generated during experiments) was provided in the room. Reagents, glassware, sampling equipments and analytical instruments were arranged in designated areas. The laboratory now has the capacity to carry out a

wide range of analysis. Laboratory was strengthened to analyze parameters viz. pH, temperature, Dissolved Oxygen, Biochemical Oxygen Demand, Chemical Oxygen Demand, chlorides, sulfate, ammonia, nitrate, phosphate, conductivity, and turbidity, according to Standard Methods of Water and Waste Water Analysis (APHA, 2000). Following instruments are calibrated and placed in proper working condition inside the lab:

1. Water Distillation Unit : Quartz make with capacity 3 liter/hr, electrically operated
2. Soxhlet Extraction Mantle with Digestion Unit : Six mantel type, maximum temperature 100 °C
3. B.O.D. Incubator : Digitally controlled, with temperature range 5 °C to 60 °C and accuracy + 0.5 °C
4. Hot air Oven : Range 50 °C to 450 °C, digitally Controlled
5. Muffle furnace : Digitally Controlled, maximum temperature 1150 °C
6. Magnetic Stirrer with Hot Plate : 2-liter capacity with energy regulator
7. Turbidity Meter : Range 0 to 100 NTU with 0.1 NTU resolutions and with tungsten lamp
8. Conductivity Meter : Digital display with automatic temperature compensation



Fig. 1 : EIA Laboratory at Maitri

9. pH Meter : Range 0 to 14, digital display with automatic temperature compensation
10. Digital Spectrometer : Range 340 – 960 nm, with resolution 1 nm
11. Heating Mantle : Maximum surface temperature of 450° C, controlled by a built - in energy regulator

## **METHODOLOGY**

### **Air Quality Monitoring**

The basic air quality parameters include Suspended Particulate Matter (SPM), Sulfur dioxide (SO<sub>2</sub>) and Oxides of Nitrogen (NO<sub>x</sub>) in the ambient air. Suspended Particulate Matter (SPM) in the flue gas which generates from the vents of incinerators and boilers was also measured.

#### **Suspended Particulate Matter (SPM)**

**Method :** Gravimetric analytical method adopted for the ambient air quality monitoring.

**Instrument & Materials :** High Volume Sampler, Glass fiber filter paper, analytical balance (4-decimal places), Oven.

**Procedure :** High Volume Sampler draws air from the atmosphere at a rate of 1-1.5 m<sup>3</sup>/min. This is made to pass through a glass fiber filter paper of known pore size and weight. Dust particulates collect on it during the 24 hourly run of the instrument. At the end of that period the filter paper is removed, oven-dried and weighed to estimate the SPM in the air (Thomas et al., 1972). SPM is expressed in µg/m<sup>3</sup>.

#### **Sulfur dioxide (SO<sub>2</sub>)**

**Method :** The analytical method adopted is colorimetric.

**Instrument & Materials :** High Volume Sampler, Rota meter, Spectrophotometer, cuvettes, glass impingers, impinger stands, desiccators, pipettes, beakers, rubber-tubing, KCl, HgCl<sub>2</sub>, para-Rosaniline, formaldehyde, sulphamic acid.

**Procedure :** Potassium tetrachloromercurate of known strength is used as the absorbing medium. A known volume of the medium is taken in impinger. It is attached to the High Volume Sampler via manifold and air is drawn through it at a rate of 0.5 litres/min. As the air passes through the absorbing medium, the target pollutant gets absorbed in it. After a 24 hour run the

absorbing media were treated with specified reagents in order to remove interference and to develop colour. The concentration of the  $\text{SO}_2$  is determined by measuring absorbance at 560nm on spectrometer (Thomas et al., 1972).  $\text{SO}_2$  is expressed in  $\mu\text{g}/\text{m}^3$ .

### **Nitrogen Oxides (NO<sub>x</sub>)**

**Method :** The analytical method adopted is colorimetric.

**Instrument & Materials :** High Volume Sampler, Rota meter, glass impingers, impinger stands, Spectrophotometer, cuvettes, desiccators, pipettes, beakers, rubber-tubing, hydrogen peroxide, sulphanilamide, NEDA.

**Procedure :** Sodium hydroxide & Sodium arsenite solution of known strength is used as the Absorbing medium. A known volume of the medium is taken in impinger. It is attached to the High Volume Sampler via manifold and air is drawn through it at a rate of 0.5 litres/min. As the air passes through the absorbing medium, the target pollutant gets absorbed in it. After a 24 hour run the absorbing media were treated with specified reagents in order to remove interference and to develop colour. The concentration of NO<sub>x</sub> is determined by measuring absorbance at 550nm on spectrometer (Thomas et al., 1972). NO<sub>x</sub> is expressed in  $\mu\text{g}/\text{m}^3$ .

### **Stack Monitoring**

The SPM monitoring was conducted by a portable Stack Monitoring kit. S type Pitot tube implied to measure velocity of the stack gas, whereas the temperature measured using the thermocouple. Pre-conditioned and weighted Glass fiber thimble fitted into the holder to trap the SPM from the stack. After applying necessary corrections for the temperature and pressure exact velocity was calculated. The concentration of SPM in the flue gas was calculated adopting gravimetric analysis (Thomas et al., 1972).

### **Water Quality Monitoring**

Several glacial and fresh water lakes exist around the station. The main lake known as Priyadarshini Lake is source for potable water is located in front of the station which is feed by interconnected glacial lakes. Only six locations (denoted as W1 to W6 in Figure 3) could be sampled because due to high wind sailing in the lake did not offer favorable condition to collect the water samples. In this lake six sampling spots were selected for water sampling, taking into account importance of particular segment i.e.

feeding inlet, outlets of lake, conjunction of lake and seepage channel through wastewater system, nearness to summer bathrooms etc. Samples were collected in the January 2004 when lake's top layer melted due to change in ambient temperature. Two locations in the glacial lakes (denoted as W8 and W9 in Figure 3) located nearby Maitri station were also selected for water sample collection.

Basic parameters of water and wastewater i.e. pH, temperature, conductivity, turbidity, DO, BOD, COD, alkalinity acidity, TDS, ammonia, nitrite and nitrate were analysed in Environmental laboratory at Maitri station. The preserved samples were sealed properly and brought to India for further analysis in the laboratory. Samples were filtered before analysis with 0.45  $\mu\text{m}$  Whatman filter papers. All the samples were analyzed using standard methods of water and waste water analysis for, total hardness, calcium hardness, sulfate, sodium, potassium, oil and grease, hydrocarbons, phenol and trace metals viz. nickel, cadmium, chromium, copper, lead, iron, manganese, zinc, and cobalt (APHA,2000).

The samples were analyzed according to standard methods of water and wastewater analysis as outlined below:

Turbidity by Nephelometric method, acidity by titrametric method, alkalinity by titrametric method, hardness by EDTA titrametric method, conductivity by conductivity meter, solids by gravimetric and oven dried method, sulfate by turbiditric method, temperature by mercury filled certified celsius thermometer and thermocouple, metals by atomic absorption spectrometry, chlorides by argentometric method, fluoride by SPADNS method, pH by electrometric method, nitrogen (ammonia, nitrate and nitrite) by colorimetric method, phosphorus by colorimetric method, BOD by 5 days BOD test, COD by closed reflux colorimetric and open reflux titrametric method, oil and grease by partition gravimetric method (APHA, 2000).

### **Performance Evaluation of RBC**

Gray water generated from kitchen is treated by B1 RBC, whereas gray water from other sources like bathroom, laundry, urinal and washbasin is treated from a separate B3 RBC. Treatment systems are installed and operated to filter the gray water and convert organic matter into stabilized unharmed material and gases. After treatment, effluent is collected in a collection pond. Once in a year during summer period accumulated treated

waste water of pond is discharged around 200 meters away from pond on the open ground near glacial lake. Both the treatment systems are kept outside of the main building in a separate wooden chamber adjacent to station building. Pipes used for carrying wastewater are coiled with trace heating element to keep liquid warm to avoid freezing due to very low ambient temperature.

To assess the efficacy of the treatment system, wastewater sample locations were selected based on its point of generation, treatment and discharge. Sample collection points were indentified i.e. inlet points, primary and secondary settling tanks and biozones. Sample was also collected from the collection pond. Physico-chemical, nutrient, oxygen demand and trace metal analysis of the samples were carried out to assess degree of function of treatment system.

## RESULTS & DISCUSSION

### Air Quality

Ambient air quality monitoring (AAQM) was carried out in the Schirmacher region surrounding to Maitri station. Five sample locations

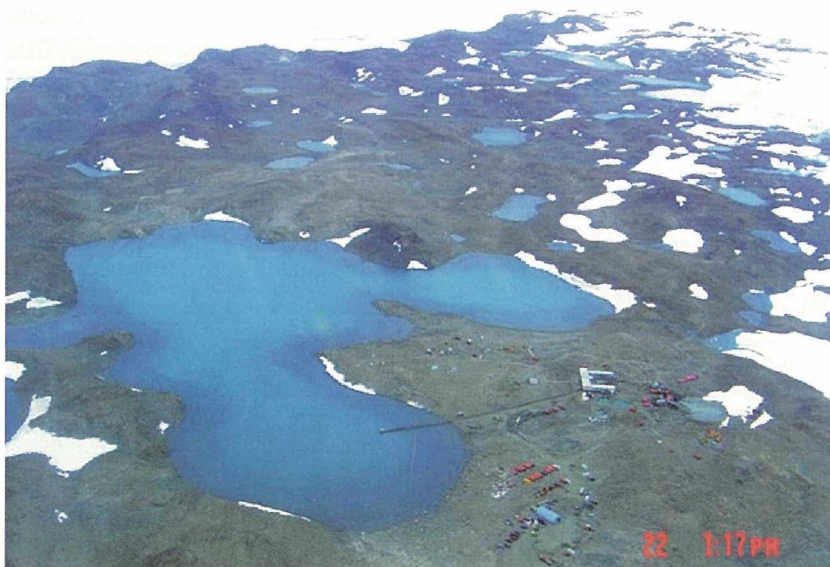


Fig. 2 : Maitri Station, Priyadarshini Lake and Summer camp

were identified for this purpose (Figure 2) around the Maitri station. These were selected in such a way that four of them were located in the downwind direction of a known source(s) of air pollution while one of the monitoring stations was located in the upwind direction of Maitri station in order to gather information on the background concentration prevailing in the ambient atmosphere. The existing sources of air pollution at Maitri station were identified viz. generators, boilers, incinerator-based toilets and exhaust emitted during vehicular movement (piston pulley, crane and dozer). It is also noticed that SPM in the atmosphere were being added by helicopter during hovering, landing and takeoff. Keeping in view the type of fuel (Aviation Turbine Fuel Jet A-1) used at Maitri station in life supporting systems as well as in vehicles, gaseous pollutants monitored were sulfur dioxide ( $\text{SO}_2$ ) and oxides of nitrogen ( $\text{NO}_x$ ). These pollutants were monitored on 24 hourly averages, continuously for a period of three to four days at each location by operating high volume air sampler (HVAS). The particulate matter was determined gravimetrically while the gaseous pollutants were estimated spectrometrically (Thomas et al., 1972).

The ambient air quality monitoring results of 24 hourly averages, obtained through running HVS within the range from 50-150 meters from the sources. Results shows that the SPM concentration around the Maitri varied from 0.5 to 4.7  $\mu\text{g}/\text{m}^3$ , the highest concentration was recorded near by the excreta incinerator and the lowest recorded at upwind direction i.e. east side of the station (Table 1). Average wind direction recorded during the study period from east to south-east direction. Russian station Novo is situated around four km in upwind direction from Maitri station which may be probable source for contributing SPM. Station uses the Aviation Turbine Fuel (ATF) which contained negligible SPM. However, due to incomplete combustion of the fuel and also the burning of the excreta in the incinerators, SPM is added to the atmosphere. The  $\text{SO}_2$  concentration in the ambient air quality varied from BDL to 13.7  $\mu\text{g}/\text{m}^3$  (Table 1). Similarly, the  $\text{NO}_x$  concentration varied from BDL to 10.1  $\mu\text{g}/\text{m}^3$ . Near the incinerator the concentrations of  $\text{SO}_2$  and  $\text{NO}_x$  could not be monitored due to freezing of the absorbent media owing to prevailing sub zero temperature. The generator consumes around 10-12 litre of ATF per hour, resulting in emission of SPM,  $\text{SO}_2$  and  $\text{NO}_x$  in the ambient air.

**Table 1 : Cumulative 98 Percentile Values of SPM, SO<sub>2</sub> and NO<sub>x</sub> in Ambient Air**

| Location               | SPM | SO <sub>2</sub> | NO <sub>x</sub> |
|------------------------|-----|-----------------|-----------------|
| Upwind of Maitri       | 0.5 | BDL             | BDL             |
| Near Summer Toilet     | 1.0 | 2.8             | 4.5             |
| Near Aaditya Generator | 1.6 | 13.7            | 10.1            |
| Near Boiler Room       | 1.4 | 3.8             | 9.3             |
| Near Incinerator       | 4.7 | -               | -               |

*All values are in µg/m<sup>3</sup>*

Stack monitoring result showed that the concentration of SPM varied from 3.3 mg/m<sup>3</sup> to 89.9 mg/m<sup>3</sup>. The highest concentration was recorded in the incinerator stack (Table 2). This is because of burning of the excreta in the incinerator and thereafter converting to ash. By this process a part of ash mixes with the stack flue gas.

**Table 2 : SPM Concentration in the Stacks**

| S.No | Stack         | Concentration in mg/m <sup>3</sup> |
|------|---------------|------------------------------------|
| 1    | Boiler        | 3.23                               |
| 2    | Generator     | 21.3                               |
| 3    | Maitri Toilet | 72.3                               |
| 4    | Summer Toilet | 89.6                               |
| 5    | Incinerator   | 68.5                               |

## Water Quality

### *Lake System*

Concerning the surface layer of Priyadarshini Lake, it was found that concentrations of various parameters viz. nutrient, demand, trace metal are mainly affected by the dynamics of ice melting. During this particular year of study due to absence of logistic support, sample collection from interior and centre of lake could not be repeated as done in the year 2000 (Tiwari, et al. 2005). Measurement of the temperature was carried out instantly at the site using calibrated alcohol based thermometer, while samples were

fixed for Dissolved Oxygen (DO) and later was estimated in the lab using Wrinkle's method as described in standard methods (APHA, 2000). Few analyses and measurements which were essential i.e. pH, conductivity, BOD, alkalinity and nutrients were carried out in the EIA laboratory. Trace metals, oil and grease and hydrocarbon analyses carried out in India.

Many of lakes are remnants of larger glacial water bodies. During the freezing condition although lakes are covered with ice, these lakes minimize the wind-driven currents and, to a limited extent, the exchange of gases between the water column and the atmosphere (Raymond et al., 2000). Priyadarshini Lake remains in complete epilimnion state during winter period and thereafter the moat gradually disappear during summer period for couple of days, thus the oxygen transfer rate increases.

The physicochemical characteristics of various water samples (Figure 3) are shown in Tables 3 and 4. The physical characteristics in terms of pH indicate it to be on the acidic side with pH ranging between 4.3-6.58 in Priyadarshini, Glacial Lake and tap water (inside Maitri station). The acidic nature of glacial lakes is due to absorbance of atmospheric  $\text{CO}_2$  at sub zero temperature while little higher conductivity with respect to Priyadarshini Lake is indicative of mixing of glacier melted water and abrasion over the moraine deposits in the glacier. The electrical conductivity ranges between 6.1-26.1  $\mu\text{S}/\text{cm}$ . The conductivity values indicates weak acid or bases of formation in the lake and less ions presence in water.

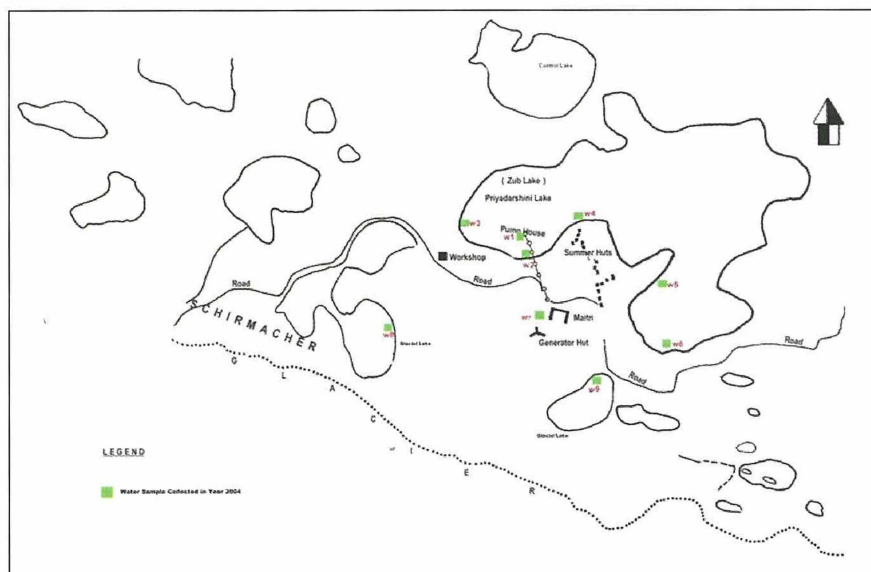


Fig. 3 : Schematic diagram of Maitri Station and Sampling Location

**Table 3 : Surface water quality - Physicochemical characteristics and Inorganic characteristics**

| Sampling Location                    | Denoted as | pH   | Temperature (°C) | Turbidity (NTU) | Dissolved Oxygen (mg/l) | Total Suspended Solids (mg/l) | Total dissolved Solids (mg/l) | Conductivity (µS/cm) | Total alkalinity (mg/l as CaCO <sub>3</sub> ) | Total Hardness (mg/l as CaCO <sub>3</sub> ) | Calcium Hardness (mg/l as CaCO <sub>3</sub> ) | Chloride (mg/l) | Sulfate (mg/l) | Sodium (mg/l) | Potassium (mg/l) |
|--------------------------------------|------------|------|------------------|-----------------|-------------------------|-------------------------------|-------------------------------|----------------------|-----------------------------------------------|---------------------------------------------|-----------------------------------------------|-----------------|----------------|---------------|------------------|
| Pump House                           | W1         | 5.6  | -1.0             | 2.0             | 14.4                    | 1                             | 16                            | 12.2                 | 8                                             | 12                                          | 6                                             | 39.3            | 2              | 3             | 1                |
| Confluence point of seepage & lake   | W2         | 4.6  | -1.0             | 1.0             | 14.2                    | 1                             | 14                            | 16.6                 | 4                                             | 8                                           | 5                                             | 43.5            | 3              | 3             | 1                |
| Confluence point of upper lake & P.  | W3         | 5.23 | -1.0             | 2.5             | 14.0                    | 1                             | 13                            | 23.6                 | 4                                             | 7                                           | 5                                             | 5.85            | 1              | 3             | 1                |
| Behind Vindhya Hut                   | W4         | 5.37 | -2.7             | 0.8             | 14.8                    | 1                             | 16                            | 8.0                  | 5                                             | 7                                           | 5                                             | 10              | 2              | 3             | 0                |
| Behind Summer Bathroom               | W5         | 6.58 | -2.7             | 2.0             | 14.8                    | 1                             | 16                            | 6.1                  | 6                                             | 11                                          | 4                                             | 32              | 2              | 2             | 1                |
| Confluence point of Glacial lake and | W6         | 5.37 | -2.7             | 1.4             | 14.8                    | 1                             | 18                            | 6.5                  | 6                                             | 13                                          | 12                                            | 16              | 1              | 3             | 1                |
| Drinking water of Priyadarshini      | W7         | 5.3  | 10.5             | 1.0             | 12.4                    | <1                            | 15                            | 7.2                  | 4                                             | 13                                          | 6                                             | 16.7            | 1              | 3             | 1                |
| Glacial Lake (R.H.S.)                | W8         | 5.4  | -1.0             | 6.0             | 14.8                    | 1                             | 15                            | 23.2                 | 4                                             | 10                                          | 8                                             | 16.7            | 1              | 3             | 1                |
| Glacial Lake (L.H.S.)                | W9         | 5.41 | -2.0             | 5.0             | 14.8                    | 1                             | 13                            | 26.1                 | 4                                             | 11                                          | 5                                             | 18.4            | 1              | 2             | 1                |

**Table 4 : Surface water quality - Organic & nutrient characteristics and concentration of selected heavy metals**

|                                                    |    |      |      |    |     |    |    |      |      |    |      |      |      |      |      |      |      |
|----------------------------------------------------|----|------|------|----|-----|----|----|------|------|----|------|------|------|------|------|------|------|
| Pump House                                         | W1 | 0.50 | 2.30 | <5 | 0.5 | ND | ND | ND   | ND   | ND | 0.01 | ND   | ND   | 0.07 | ND   | 0.02 | ND   |
| Confluence point of seepage & lake                 | W2 | 0.40 | 2.0  | <5 | 4.0 | ND | ND | 24.5 | ND   | ND | 0.01 | 0.01 | 0.03 | 0.14 | 0.01 | 0.01 | 0.01 |
| Confluence point of upper lake & P. Lake           | W3 | ND   | 2.50 | <5 | Nil | ND | ND | ND   | ND   | ND | 0.01 | ND   | ND   | 0.17 | 0.01 | ND   | ND   |
| Behind Vindhya Hut                                 | W4 | ND   | 4.20 | <5 | 3.0 | ND | ND | ND   | ND   | ND | 0.01 | 0.01 | 0.03 | 0.06 | 0.01 | ND   | 0.01 |
| Behind Summer Bathroom                             | W5 | ND   | 1.4  | <5 | 3.3 | ND | ND | ND   | ND   | ND | 0.01 | ND   | ND   | 0.18 | ND   | ND   | ND   |
| Confluence point of Glacial lake and Priyadarshini | W6 | ND   | 6.30 | <5 | Nil | ND | ND | ND   | ND   | ND | 0.01 | 0.01 | ND   | 0.09 | ND   | ND   | ND   |
| Drinking water of Priyadarshini                    | W7 | 0.31 | 0.04 | <5 | Nil | ND | ND | ND   | 0.03 | ND | ND   | 0.39 | ND   | 0.08 | ND   | 0.03 | ND   |
| Glacial Lake (R.H.S.)                              | W8 | 0.40 | 1.80 | <5 | 5.6 | ND | ND | ND   | ND   | ND | 0.01 | ND   | ND   | 0.20 | 0.01 | ND   | 0.01 |
| Glacial Lake (L.H.S.)                              | W9 | ND   | 6.30 | <5 | 4.3 | ND | ND | ND   | ND   | ND | ND   | ND   | ND   | 0.19 | ND   | ND   | ND   |

*All values are in mg/l except hydrocarbon, ND-Not Detectable*

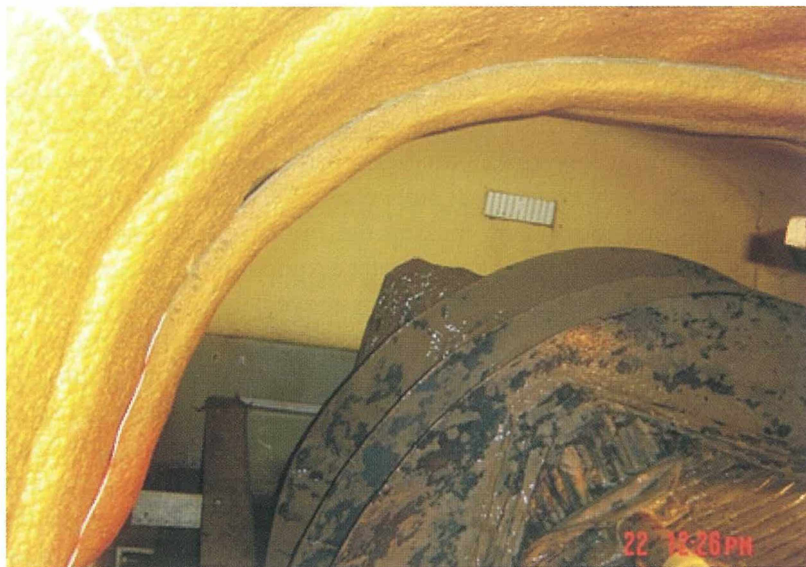
The inorganic parameters in terms of sulfate and chloride were found to be in the range of 1-3 mg/L and 10-43.5 mg/L, respectively. The nutrient parameters such as nitrate and phosphate were recorded to be in the range of 0-0.5 mg/L. Dissolved oxygen level in various water samples was found in between 12.4-14.8 mg/L while temperature at these sampling locations varied from -2.7°C to 10.5°C, indicating a very high re-aeration capacity at the surface of water bodies. The concentrations of chloride, nitrates, BOD<sub>5</sub> and COD in the Priyadarshini Lake water were found to be well within permissible limits, but chloride level was on a higher side at the confluence point of wastewater seepage channel and of Priyadarshini Lake. This might be due to incoming seepage and decomposition of moss which was grown surrounding to lake. The mixing of wastewater produce from the bathroom and urinal into lake is being contributed by inorganic phosphate which contributed by breakdown of proteins and elimination of the librated phosphates in the urine and other organic and inorganic substances responsible for BOD and COD. It was also observed that the values of BOD<sub>5</sub>, conductivity and chlorides in the glacial lake lying on the western side of station were on the higher side as compared to that on the eastern side. This might be owing to the fact that the former water body is located at a lower contour at a distance of around 200 m from the point of discharge of partially treated effluent during the summer period. The possibility of contamination of glacier water through seeped wastewater cannot be ruled out. Trace metal levels in all the water samples were recorded well within permissible limits of drinking water. Oil and grease and hydrocarbon were not observed in any of the water samples collected from the Lakes.

## **WASTEWATER TREATMENT**

### **Performance Evaluation of RBC**

The wastewater treatment (RBC) systems installed at Maitri station during year 1989-90. These RBCs were designed and manufactured by Klargester Company, UK. Since beginning the wastewater produced at Maitri is segregated and are being treated in different RBCs to treat the gray water generated in station. They are made of Glass fiber Reinforced Polyester (GRP) which is extremely robust and easy to transport and install. The RBC units are located in separate rooms that are not exposed to sunlight to prevent the growth of algae, which could otherwise intervene with microbial attachment (Ayoub et al., 2004) and at times overburden the discs with extra mass. The RBC comprises main components such as: primary settling zone, biozone and secondary settling zone. The direction of disc rotation is

selected to be opposite to the direction of wastewater flow in order to reduce on short circuiting (Fig. 4).



*Fig. 4 : RBC, Showing Rotating Disk*

### **Kitchen RBC (B1)**

Kitchen waste contains high organic matter which is originated from washing of food grains, dishes, condensed milk waste and vegetable oil waste and other sources. It is also observed that the strength of the waste changes day to day as per the food habits and number of person present at Maitri station. Hydraulic loading on the RBC found maximum during summer as the strength of member increases. During study period and collection of samples it was recorded  $0.08 \text{ l/m}^2\text{d}$ . The composite samples collected from inlet, outlet and other places on a particular day when strength was maximum at station and variety of food served during particular occasion. So it imparted higher organic as well as hydraulic load on RBC. The reason behind this was to assess the treatment efficacy of RBC during maximum fluctuation and shock loading conditions. The average water consumption at Maitri station recorded was 3.0 kl, out of which 1.045 kl is used in kitchen. After being consumed for drinking, cooking and evaporation loss wastewater produced and reaches to Kitchen RBC (B1). Volume of wastewater measured approximately 0.7 kl per day during the summer period from the outlet of the kitchen RBC. However during winter period

water consumption reduces, so wastewater production is also reduces accordingly.

Analysis shows that wastewater is of acidic nature and septic which contains high strength of BOD and COD. Due to lower pH, the ammonia nitrogen of effluent is higher than the influent, which indicate that RBC is stabilizing the organic matter but the ammonia producing bacteria are not acting effectively at lower pH. Nitrification is probably hampered due to low alkalinity in the wastewater (Gupta et al., 2001). Oil and grease which is also mixing in the influent, making the film over the liquid and thus disrupting the oxygen transfer from surface to bottom of tank however in the biological chamber due to continuous movement of the disc, oil film cannot formed rather it is emulsifying the liquid. Microbial growth was observed on the disc but of grey color which is clear indication of the overloading on the treatment system. The biomass development in aerobic unit did not always cover the entire disk surface (Hiras et al., 2004). Mostly the BOD and COD are removed in the sedimentation process in the primary and secondary settling chambers of tanks and biological chamber resulting reduction of the BOD and COD load which is mostly imparted due to dissolved matter.

Few trace heavy metals are removed very effectively by system. The reason behind is that copper, iron and zinc which are essential growth elements for bacteria are consumed by the bacteria. The temperature sometimes maintains at biodisk to 20-22°C (Table 5). The BOD removal efficiency was observed to be 48.3% and COD to be 24.7% while organic load was higher side. This RBC was designed for maximum uptake of BOD of 300 mg/l in the influent with the maximum flow of 1.32 m<sup>3</sup>/day. High BOD and COD were adding to the effluent from the secondary settling tank from where the sludge was not removed properly. Due to thick sludge formation anaerobic condition occurred which produced gases and resulted in keeping sludge particles in suspension. More or less this process adduces to add BOD and COD up to the effluent. Oil and grease and hydrocarbon are also responsible to hamper the treatment efficiency of the system. Among the heavy metals in the sample copper, manganese, zinc concentrations were found to be lower in effluent except the iron content (Table 6).

### **Urinal-Laundry-Bathroom RBC (B3)**

Another RBC (B3) was also installed in the year 1989-90 at Maitri which is designed and manufactured by Klargestor Company, UK (Kee,

2004). This RBC is being used to assimilate the waste generated from bathroom, urinal and laundry. Hydraulic loading on the RBC during summer recorded as  $0.102 \text{ l/m}^2\text{d}$ . The average water consumption for bathroom, urinal and laundry recorded around  $2.0 \text{ kl}$  in a day and wastewater produced that reaches RBC (B3) is approximately  $1.95 \text{ kl}$  per day during the summer period. The samples were collected in the first week of January 2004. Analysis result shows that at inlet wastewater pH is normal in nature, DO content is very less (Table 5), however moderate strength of BOD and COD were recorded. The treatment efficacy is observed to be only 16.6% of BOD and 25.7% of COD. This RBC was designed for maximum uptake of  $300 \text{ mg/l}$  of BOD in the influent with the maximum flow of  $5.0 \text{ m}^3/\text{day}$ . BOD load is less than the design load and the pH of the effluent is recorded of reasonable quality. Laundry effluent consists of, nitrogen and phosphorus required for bacteria cell reproduction but it is noticed through analysis of samples that carbon, calcium, magnesium and other trace elements are not present in the effluent to reproduce the bio cells. This adduces less production of microbes on biodisk which is ultimately affecting the treatment efficiency. Oil and grease and hydrocarbon are also responsible to hamper the treatment efficiency of the system. Among the heavy metals in the sample copper, manganese and zinc concentrations are found to be lower in the effluent except the iron content (Table 6).

### **Modular Design**

Plants must also be able to handle widely varying hydraulic and BOD loads, since number of people wintering at research stations is less than during the peak summer resupply period. Most people present on the bases have similar work schedules, so diurnal load fluctuations are pronounced and have to be allowed. Another aspect is that numbers on stations tend to alter stepwise, for example when the resupply vessel arrives. This means that plants need to be adjusted rapidly to the load changes. This problem is most acute at seasonally occupied stations.

The climate, together with the prolonged period of darkness, during winter and the fact that stations maintain only a skeleton staff during the winter months, means that there exists a total of perhaps eight weeks in summer when conditions are good for outdoor activities (Heaton et al., 2003). Much of this time will be needed for transport of plant components to the selected site, installation, and connection of pipes and utilities. Therefore modular, prefabricated plants that can be easily disassembled for shipping and readily reassembled on site are therefore virtually a necessity. Such

Table 5 : Wastewater Quality (Physical and Inorganic Parameters)

| Sr. No. | Sampling Location         | Temperature ( $^{\circ}\text{C}$ ) | pH   | Dissolved Oxygen (mg/l) | Turbidity (NTU) | Total Solids (mg/l) | Total Suspended Solids (mg/l) | Conductivity ( $\mu\text{S}/\text{cm}$ ) | Total alkalinity (mg/l) | Chloride (mg/l) | Sulfate (mg/l) |
|---------|---------------------------|------------------------------------|------|-------------------------|-----------------|---------------------|-------------------------------|------------------------------------------|-------------------------|-----------------|----------------|
| 1.      | WW1- Inlet Kitchen        | 22                                 | 4.99 | 10.4                    | 200             | 1188                | 18                            | 207                                      | 22                      | 20              | 24             |
| 2.      | WW2- Outlet Kitchen       | 21                                 | 4.54 | 1.0                     | 190             | 131                 | 18                            | 202                                      | 81                      | 28.4            | 5              |
| 3.      | WW3- Inlet Urinal         | 21                                 | 6.98 | 0.54                    | 55              | 291                 | 1                             | 849                                      | Acidic                  | 87              | 300            |
| 4.      | WW4- Outlet Urinal        | 21                                 | 7.28 | 4.6                     | 42              | 51                  | 6                             | 941                                      | 226                     | 95              | 34             |
| 5.      | WW5- Wastewater pond      | 1.5                                | 7.67 | 4.6                     | 85              | -                   | 13                            | 1053                                     | 251                     | 128.8           | 31             |
| 6.      | WW6- Biochamber – Kitchen | 20                                 | 7.1  | -                       | -               | -                   | 6                             | -                                        | 167                     | 25              | 26             |
| 7.      | WW7- Biochamber – Urinal  | 19                                 | 8.4  | -                       | -               | -                   | 9                             | -                                        | 300                     | 97              | 42             |

**Table 6 : Waste Water Quality (Nutrient, Demand and Organic Parameters and Heavy Metals)**

| Sr. No. | Sampling Location        | Nitrate as N | Ammonia as N | Total Phosphates | Chemical Oxygen Demand | Bio-chemical Oxygen Demand | Oil and grease | Hydro carbons ( $\mu\text{g/l}$ ) | $\text{NO}_2\text{-N}$ | $\text{NH}_3\text{-Nitrogen}$ | Phenol | Nickel | Cadmium | Chromium | Copper | Lead | Iron | Manganese | Zinc | Cobalt |
|---------|--------------------------|--------------|--------------|------------------|------------------------|----------------------------|----------------|-----------------------------------|------------------------|-------------------------------|--------|--------|---------|----------|--------|------|------|-----------|------|--------|
|         |                          | mg/l         |              |                  |                        |                            |                |                                   |                        |                               |        |        |         |          |        |      |      |           |      |        |
| 1.      | WW1- Inlet Kitchen       | 0.10         | 285          | 2.10             | 728                    | 620                        | 3.9            | 0.44                              | 0.05                   | 4.70                          | 0.13   | ND     | ND      | ND       | 0.02   | ND   | 0.14 | 0.01      | 0.05 | ND     |
| 2.      | WW2- Outlet Kitchen      | 0.13         | 354          | 11.10            | 548                    | 320                        | 2.1            | 0.30                              | 0.04                   | 71.50                         | 0.25   | ND     | ND      | ND       | 0.05   | 0.06 | 0.84 | 0.02      | 0.09 | 0.02   |
| 3.      | WW3- Inlet Urinal        | 0.12         | 980          | 17.40            | 280                    | 120                        | 2.6            | 0.30                              | 0.01                   | 71.0                          | 0.16   | 0.14   | ND      | ND       | 0.12   | 0.05 | 0.17 | 0.01      | 0.10 | 0.01   |
| 4.      | WW4- Outlet Urinal       | 0.13         | 2180         | 12.40            | 208                    | 100                        | 4.0            | 0.27                              | 0.01                   | 165.0                         | 0.15   | 0.04   | 0.12    | ND       | 0.24   | 0.05 | 0.13 | 0.01      | 0.09 | ND     |
| 5.      | WW5- Wastewater pond     | 0.17         | 1525         | 17.20            | 452                    | 260                        | 5.2            | 0.28                              | 0.01                   | 58.0                          | 0.05   | 0.03   | ND      | ND       | 0.03   | 0.06 | 0.39 | 0.10      | 0.04 | 0.01   |
| 6.      | WW6- Biochamber – Urinal | 0.14         | -            | 13.8             | 1536                   | 675                        | 1.3            | 0.16                              | 0.01                   | 65.0                          | 0.46   | 0.16   | ND      | 0.01     | 0.26   | 0.07 | 1.70 | 0.05      | 0.63 | 0.04   |
| 7.      | WW7- Biochamber – Urinal | 0.15         | -            | 14.40            | 812                    | 210                        | 1.8            | 0.21                              | 0.01                   | 158.0                         | 0.32   | 0.20   | ND      | 0.01     | 0.13   | 0.05 | 0.38 | 0.01      | 0.13 | 0.01   |

ND: Not detectable

plants can also be commissioned and modified prior to shipping, alleviating demands on the limited staff available on-site at research stations. For convenience of transport and handling, and to protect plant components during the rough weather characteristics of the southern ocean, most countries are opting to build their plants inside the containers. Plants need also to have low energy consumption. Energy is expensive in Antarctica, being derived from hydrocarbon fuels shipped in at high cost and stored in heated storage tanks. Aside from saving money, installing a low power demand plant means a reduced output of emissions from the base's power station (Connor, 2008).

The Scott base plant fits inside two ISO-20 shipping containers (New Zealand, 2002), while, the Neumayer and Jubany plants were built into containers 2.0 m long, 1.8 m wide and 1.8 m high, made of stainless steel to minimize problems with corrosion (Ahammer et al., 2000). Compactness is necessary if plants are to fit into containers. It also has other advantages, reducing the plant's footprint and its visual impact (Heaton et al., 2003) as well as minimizing heating costs. Rate of microbial waste decomposition processes are temperature dependent and till date plants have all been designed to run at 12°C (New Zealand, 2002) or above. Similarly a modular treatment system can be installed at Maitri station to meet the objectives

## **CONCLUSION**

Both the RBC's are designed to accept the BOD load of 300 mg/l at optimum temperature. The BOD load from the kitchen waste is very high, which contains high organic matter. The rpm is also recorded low which should be around 5. The reduced rpm transfers less oxygen to the bacteria and diffuses less oxygen to the liquid inside the biozone. Even though the RBCs are designed to bear the shock loading, but due to prevailing low ambient temperature and fluctuation in the flow the treatment process is affected. This statement is justified observing that ammonia removal rate significantly decreased by sudden cooling. It is possible to improve the efficacy of the RBC if system can be changed to step feed from the present single feed mode. Waste water obtained from the different sources is being treated in two different RBC's. This is resulting in availing more food and trace elements in B3 RBC, which are essential in cell growth of bacteria. The maximum flow of B3 RBC is 5 m<sup>3</sup>/day and the overall waste flow of Maitri is around 3.1 m<sup>3</sup>/day, so all the waste can be treated into single RBC (B3) if after treatment, B1 RBC effluent is mixed at the inlet of B3 RBC along with pre-designated influent of B3 RBC. This will not only neutralize the pH of waste but also provide essential growth elements to bacterial

cell. Bioaugmentation of nitrifying bacteria for short solids retention time nitrification is an attractive alternative for wastewater treatment plants in cold climate and mixing of centrate from kitchen RBC can be used as a source for bioaugmentation of B3 RBC. Oil and grease trap should be fixed before the waste enters into RBC which will help in faster cell development rather than becoming toxic to bacteria.

The environmental assessment exercise throws light on the status of air and water in and around Maitri station. The temperature inside the RBC should be maintained between 12 and 20°C throughout the year effectively for denitrification process. It is observed that at Maitri station the environmental management is being followed according to the Madrid Protocol. Despite all care there is always scope for accidental contamination as well as transport of some solid waste into lakes due to strong winds which can directly or indirectly affect the quality of water and soil in isolated pockets albeit to a very small extent. The water quality of the lake has not changed by the intrusion of small quantity of wastewater through seepage water channel probably due to high dilution ratio maintained and also due to replenishment of water from the glacial melt (Tiwari et al, 2006). All said and done preserving the Antarctic wilderness in its original form is man's first goal. Modular design of the treatment plant can be introduced to improve efficacy by maintaining the working temperature of system and easy maintenance.

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